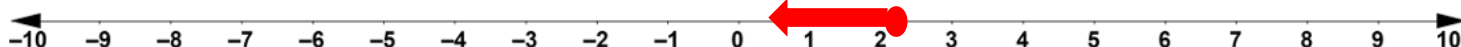


1. $3.8x - 24.46 \leq 2.3(4 - 5x)$

work

Solution

$x \leq 2.2$

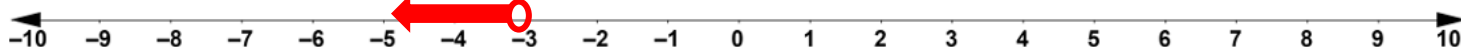


2. $-\frac{3}{4}(19 + 7x) - \frac{13}{4} > \frac{7}{4}(2x + 5)$

work

Solution

$x < -3$



3. $-1 < -3x + 11 \leq 20$

work

Solution

$-3 \leq x < 4$



4. A biologist records the average weight of the brain and body for a number of mammal species. The biologist uses the inequality $60 \leq 0.5x + 2.5 \leq 250$ to determine the weight of the mammals whose brain weighs between 60 pounds (lb) and 250 lb. Solve the inequality.

Solution

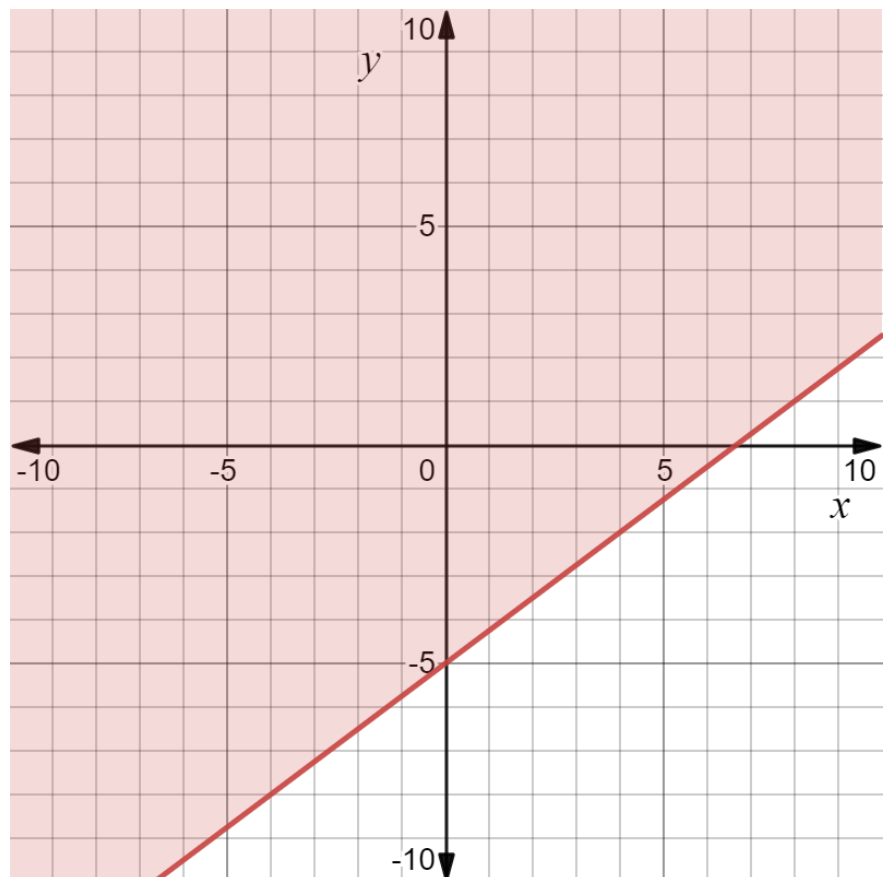
$$115 \leq x \leq 495$$

5. The smallest size of a hybrid bike, a bike that can be ridden both on and off dirt, is 16.9 inches (in.). Bikes are fitted using a person's inseam with the smallest inseam for a hybrid bike being 25 in. For inseams 25 inches or longer, each 0.68 of an inch increase in bike size, a person's inseam is 1 inch longer. A medium size bike has a range of 20.9 in. to 22 in. Write an inequality, where x is the length of an inseam, in inches, that can be used to determine the inseam sizes that best fit a medium size bike.

Inequality

$$20.9 \leq 0.68(x - 25) + 16.9 \leq 22$$

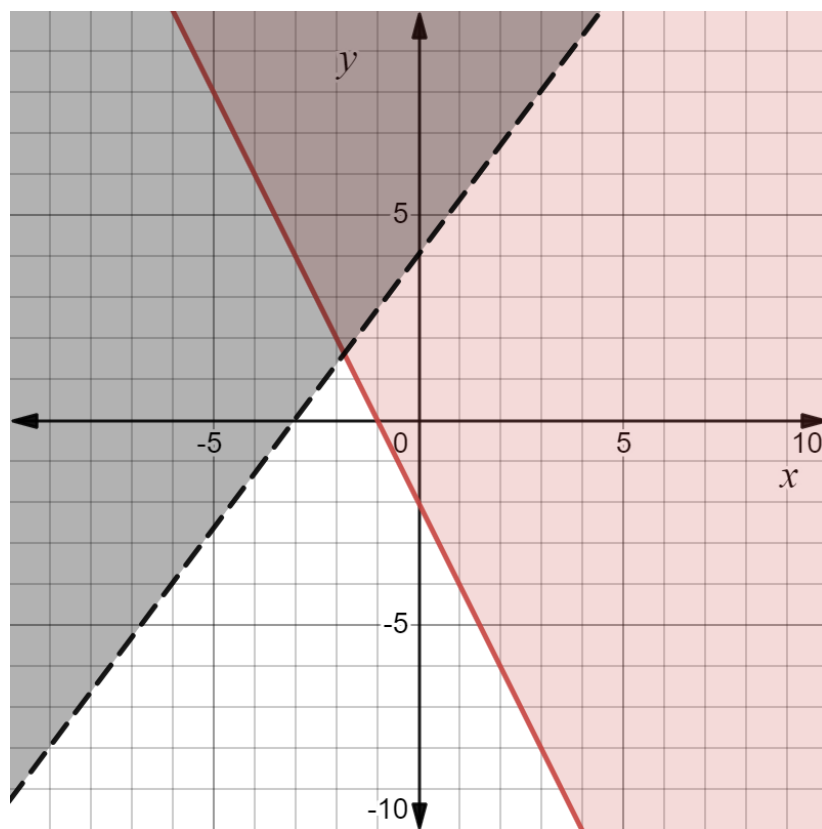
6. Graph the solution to the inequality $y \geq \frac{3}{4}x - 5$.



7. Graph the solution to the system of inequalities.

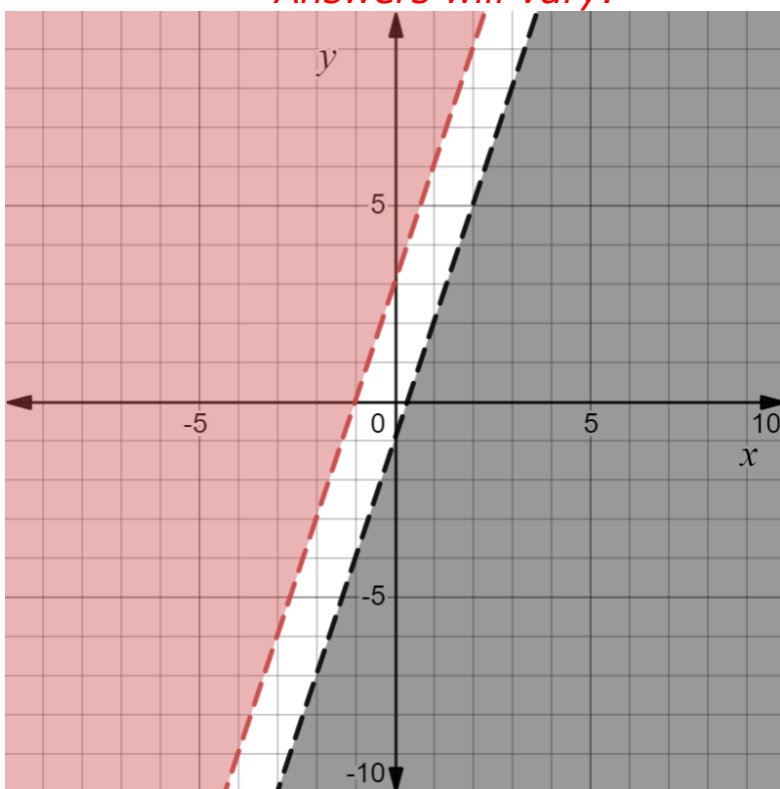
$$y - 4 \geq -2(x + 3)$$

$$4x - 3y < -12$$



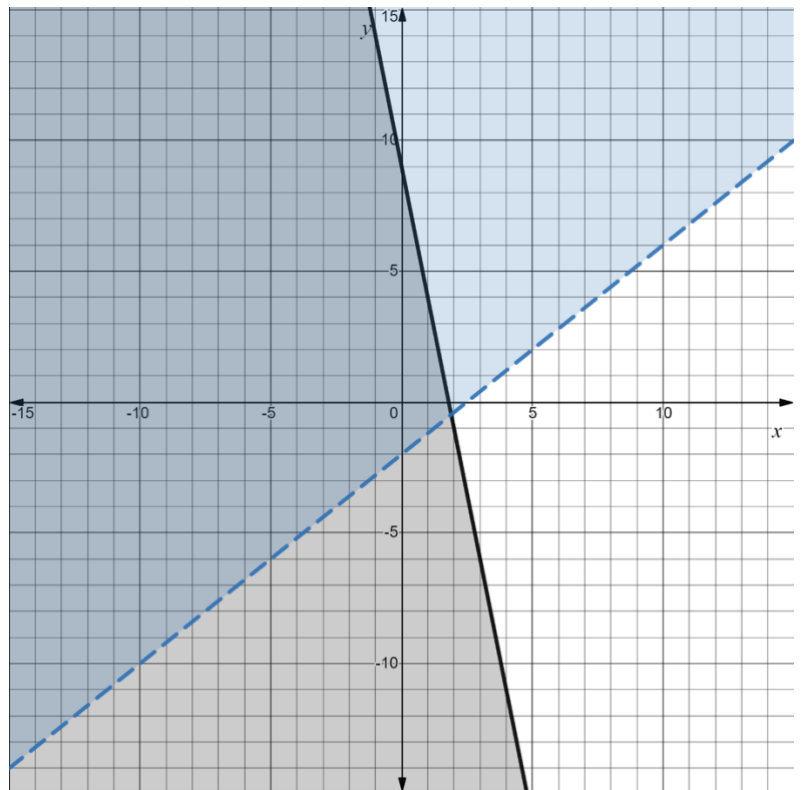
8. Draw a system of inequalities that would have no solution.

Answers will vary.



9. A system of inequalities is shown on the coordinate grid.

Write the system of inequalities that is represented by the graph.



System of Inequalities

$$\begin{aligned} y &\leq -5x + 9 \\ y &> \frac{4}{5}x - 2 \end{aligned}$$

10. To'ak Chocolate is the rarest chocolate in the world with a cost of \$5 per gram. Saffron is a spice used for flavoring and coloring foods and costs \$10 per gram. A party planner for a celebrity event wants to include no more than 5 grams (g) of To'ak Chocolate and Saffron in a gift bag. She has to keep the cost to less than \$35.

a. Write a system of inequalities where x is the number of grams of To'ak Chocolate and y is the number of grams of Saffron in each gift bag.

System of Inequalities

$$\begin{aligned} x + y &\leq 5 \\ 5x + 10y &< 35 \end{aligned}$$

b. Graph the system of inequalities.

